

Acid Erosion of Carbonate Fractures and Accessibility of Arsenic-Bearing Minerals: *In Operando* Synchrotron-Based Microfluidic Experiment

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Cite This: *Environ. Sci. Technol.* 2020, 54, 12502–12510



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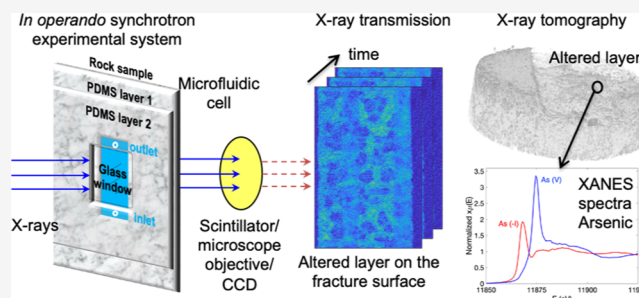


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ABSTRACT: Underground flows of acidic fluids through fractured rock can create new porosity and increase accessibility to hazardous trace elements such as arsenic. In this study, we developed a custom microfluidic cell for an *in operando* synchrotron experiment using X-ray attenuation. The experiment mimics reactive fracture flow by passing an acidic fluid over a surface of mineralogically heterogeneous rock from the Eagle Ford shale. Over 48 h, calcite was preferentially dissolved, forming an altered layer 200–500 μm thick with a porosity of 63–68% and surface area $>10\times$ higher than that in the unreacted shale as shown by xCT analyses. Calcite dissolution rate quantified from the attenuation data was $3 \times 10^{-4} \text{ mol/m}^2\text{s}$ and decreased to $3 \times 10^{-5} \text{ mol/m}^2\text{s}$ after 24 h because of increasing diffusion limitations. Erosion of the fracture surface increased access to iron-rich minerals, thereby increasing access to toxic metals such as arsenic. Quantification using XRF and XANES microspectroscopy indicated up to 0.5 wt % of As(-I) in arsenopyrite and 1.2 wt % of As(V) associated with ferrihydrite. This study provides valuable contributions for understanding and predicting fracture alteration and changes to the mobilization potential of hazardous metals and metalloids.



1. INTRODUCTION

Fractures provide preferential pathways for fluid flow and solute transport in many Earth and environmental systems. When acidified fluids, whether naturally occurring or resulting from subsurface engineering, flow through host rocks composed of relatively soluble minerals (e.g., carbonates), fracture surfaces can undergo alterations.^{1,2} These alterations are critically important to a number of properties including permeability, diffusivity, and reactive surface area.^{3–5} Consequently, these processes can impact the leakage risks of caprocks overlying storage targets for CO_2 and nuclear waste, transport and fate of contaminants in groundwater, and production rates of geothermal fields and shale gas drilling.^{6–8}

In single-mineral rocks, the fracture alteration is related to the relative magnitudes of mass transfer and reaction rate.⁹ In multimineral rocks in which fast-reacting minerals are distributed within a relatively insoluble matrix, geochemical reactions leave behind a porous altered layer.^{10–14} Fracture flow experiments coupled with X-ray computed tomography (xCT) have provided physical insights regarding altered layer development¹⁰ and enabled quantification of altered layer thickness and the modeling of its evolution.^{4,5,15} This alteration can expose slow-reacting minerals¹⁰ while imposing a diffusion limitation on the fast-reacting mineral(s).^{4,10} The altered layer typically has weakened mechanical properties that

may result in particle detachment and affect fracture morphology and stiffness under stress.^{5,13,16,17}

Development of the altered layer may have important implications for mobilization of chemicals in the rock matrix. For instance, interactions between hydraulic fracturing fluids and Eagle Ford shale samples were found to increase pyrite mobilization by 1 order of magnitude because of particle detachment following carbonate dissolution.¹⁸ Given that toxic elements, especially arsenic, are typically constituents within iron sulfides and sorbates on iron oxy-hydroxides,^{19,20} acid erosion of fracture surfaces may increase arsenic mobilization, which is an environmental concern in many subsurface practices.^{21–25}

Fracture alteration has traditionally been studied using fractured core experiments coupled with xCT imaging. These experiments provided data about fracture morphological changes and the evolution of bulk chemistry and hydraulic and mechanical properties.^{5,10,11,13} There are, however, experimental complications that arise from the cylindrical

Received: June 8, 2020

Revised: August 25, 2020

Accepted: August 26, 2020

Published: August 26, 2020



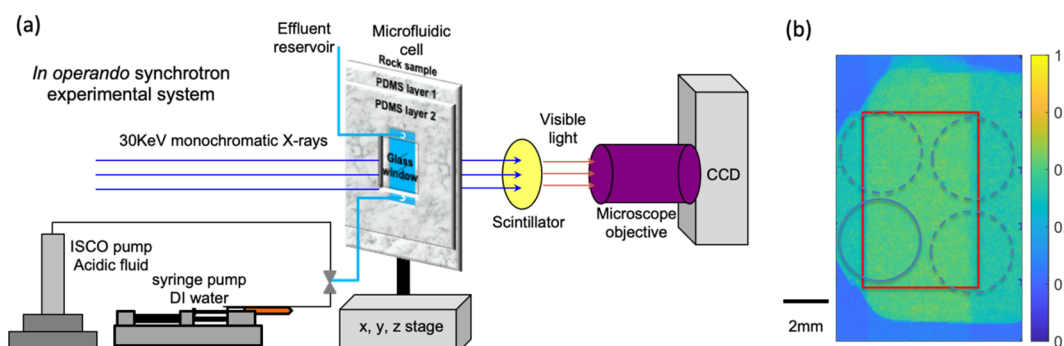


Figure 1. (a) Schematic of the microfluidic cell coupled with full-field X-ray attenuation beamline configuration and (b) transmittance (I_t) through the rock wafer at the beginning of the experiment. The hexagonal shape in the color map outlines the glass window, and the region with slightly higher transmission corresponds to the flow channel. The red box highlights the section of the channel being analyzed in the following discussions. The purple circles indicate the locations of the postmortem subcores imaged using xCT, and the solid circle in the lower left highlights the subcore shown in Figure 2.

geometry, core holders, and so on, limiting the use of the data for scientific discovery and reactive transport modeling. In this study, we used a novel approach that couples a microfluidic device with real-time synchrotron-based observations of fracture surface evolution. Microfluidic devices coupled with high-resolution cameras and a confocal microscope, initially developed and widely used for fluid mechanics and biology,^{26,27} have been adapted to study flow and reactions in geological materials.^{28,29} Recent developments, including coating the base materials with minerals^{30,31} or fabricating the microfluidic device using real rock samples,^{14,32} have made experiments involving complex chemical systems possible. Microfluidic experiments have also become an integral component in recent development and benchmarking of high-fidelity models.³³ Specifically, our experimental system has the following characteristics compared to traditional core experiments. The microfluidic cell was small enough to be accommodated in a synchrotron beamline endstation. The thin geometry made it suitable for the use of full-field X-ray attenuation, which does not require sample rotation as in xCT imaging and thus enables fast data collection. The thin geometry maximized the signal from the fracture–matrix interface, minimizing excessive X-ray attenuation that happens for thick cores. The microfluidic design did not have any compressive forces on the fracture surface, whereas compressive stress is typically unavoidable in core flow experiments because confining pressure is needed to prevent flow between the sample and the core holder. Finally, mineral dissolution rates could be measured at high temporal resolution by real-time observation of changes in the solid phase, as opposed to requiring effluent chemistry analyses as is often done in core flow experiments.

This study is an investigation of acid erosion of a rock surface with mixed mineral composition. One objective was to demonstrate the application of this real-rock microfluidic cell coupled with synchrotron-based observations. The novel cell design enabled *in operando* observation of X-ray attenuation to quantify altered layer development at high spatial and temporal resolution and quantification of carbonate reaction rates over time. The study also included postmortem xCT, X-ray fluorescence (XRF) and X-ray absorption near edge structure (XANES) microspectroscopy, and X-ray diffraction (XRD) for detailed analysis of the pore space, elemental composition, and arsenic oxidation state. In total, four synchrotron beamlines were employed to generate a rich dataset. This study

contributes knowledge of geochemical and textural alterations of fractures and discusses the implications for hydrodynamic processes and mobilization of hazardous trace elements such as arsenic. Our experimental study provides a unique dataset for future modeling of fracture surface processes and arsenic accessibility.

2. MATERIALS AND METHODS

2.1. Materials and Microfluidic Cell Fabrication. Eagle Ford shale samples were obtained from quarry blocks (Kocurek Industries) and showed the characteristically high calcite content as has been reported for this shale.^{13,18} Powder XRD (PXRD) analyses indicated approximately 70 wt % of calcite and the remaining 30 wt % originating from quartz, pyrite, and clay minerals (see the [Supporting Information](#) for PXRD patterns). Rectangular wafers (10 × 20 mm) were cut parallel to the bedding planes to prevent disaggregation and then polished to a uniform thickness of 1.5 mm. This thickness was selected to optimize the X-ray transmission signal and maintain mechanical stability over the course of the experiment.

The microfluidic flow cell was constructed by adapting established methods.²⁷ One side of the wafer was spin coated with polydimethylsiloxane (PDMS). After curing, a channel was carved out to expose the rock surface. The channel was 4.5 mm wide, 1 mm deep, and 15 mm long. Another PDMS layer was prepared with a high purity glass window for X-ray transmission. The two PDMS layers were bonded together after plasma treatment. Inlet and outlet ports were cored through the second PDMS layer, and the port-tubing connections were sealed with epoxy. The parallel plate design of the flow cell enabled a homogenized flow field. (Plan and side view of the microfluidic cell can be found in the [Supporting Information](#).)

A hydrochloric acid solution at pH 2.2 was selected as an analogue to acidified waters introduced in subsurface processes. The ionic strength was 0.1 M, which is comparable to that of moderately saline formation water in subsurface systems. It was selected to maximize the X-ray attenuation contrast between the shale and the fluid.

2.2. X-ray Attenuation Flow Experiments. Full-field X-ray attenuation measurements were used to record changes in shale sample thickness during the experiment. The X-ray tomography beamline X2B at the National Synchrotron Light Source (NSLS) was operated as a full-field X-ray microscope

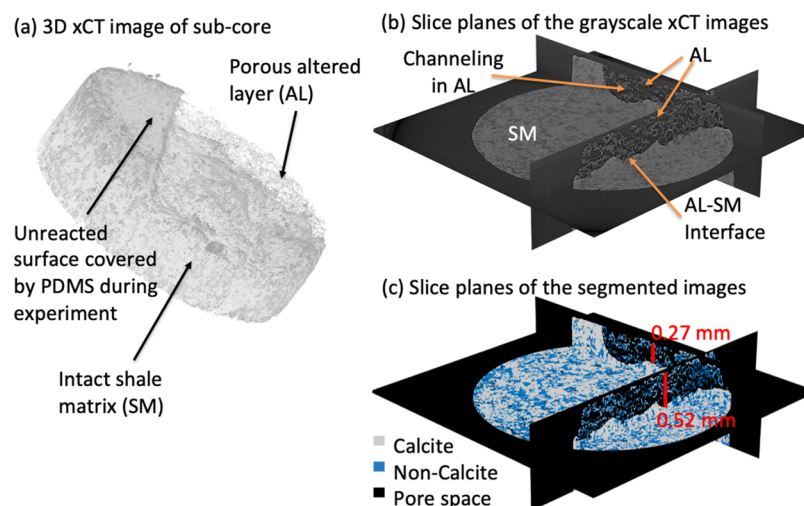


Figure 2. (a) 3D rendering of the xCT images of the postreaction subcore highlighted in Figure 1, (b) slice planes of the grayscale xCT images, (c) slice planes of the segmented images with the pore space, calcite, and the noncalcite phase colored in black, light gray, and blue, respectively. The red lines span local thickness of the altered layer.

with a monochromatic 30 KeV incident X-ray beam (details of the configuration can be found in the [Supporting Information](#)). Figure 1a shows a schematic of the microfluidic cell with the flow system mounted at the beamline. The microfluidic cell was flushed with DI water first. The acidic fluid was injected from bottom to top of the channel and was maintained at 0.2 mL/min (linear average velocity of $\sim 7 \times 10^{-4}$ m/s) with a syringe pump (Teledyne ISCO) for 48 h. This flow rate is within the range used in previous experimental investigations and could be expected for fracture flow in the subsurface.^{14,18}

A total of 15 overlapping mosaic 2D images, with a spatial resolution of 1.3 μm , of the windowed portion of the channel were collected every 15 min (Figure 1b) and the full-field images at each time point were stitched together. Data processing included the flat-field correction and the calculation of the transmittance (see details in the [Supporting Information](#)).

The transmittance (I_t) is the ratio between the transmitted X-ray intensity (I) and the incoming beam (I_0) and is an inverse measure of the attenuation by the materials in the X-ray path. For our experiment, the initial state can be described as

$$\ln(I_t^0) = A - L_w\mu_w - L_r(\mu_{nc}f_{nc} + \mu_c f_c + \mu_w \phi) \quad (1)$$

where A is a constant accounting for the transmission through materials other than the rock sample and the flow channel, for example, the glass window. Other parameters in eq 1 are the attenuation coefficients (μ), rock porosity (ϕ), and the thicknesses of the water channel (L_w) and the rock sample (L_r), respectively. For this analysis, the mineral content of the shale is represented as calcite ("c") plus all other minerals lumped together as the noncalcite ("nc") phase, with a volume fraction of

$$f_{nc} = 1 - f_c - \phi \quad (2)$$

The volume fraction of calcite, f_c , was determined in postmortem xCT analysis. During the second half of the flow experiment, periodic discontinuities in the transmission data resulting from synchrotron ring refills and orbit corrections could not be corrected. Therefore, we used the data of the second half of the experiment only for semiquantitative spatial

analyses and we limited quantitative rate calculations to data from the first half of the experiment.

2.3. XRF Measurements. Unreacted specimens of the Eagle Ford shale were examined using X-ray fluorescence microspectroscopy (μXRF) imaging to examine spatial distributions of iron and arsenic. The μXRF elemental maps of prepared thin sections were acquired at the XFM beamline (4-BM) at the National Synchrotron Light Source II (NSLS-II). A 0.5×0.7 mm region of interest (ROI) was imaged at 5 μm step size. To distinguish arsenic oxidation states, K-edge micro-X-ray absorption near edge-structure (μXANES) spectra were collected for 20 arsenic hotspots within the ROI. XANES probes the region of the spectrum within 30–50 eV of the absorption edge of an element and provides information about its oxidation state and mineral-phase association. The spectra were processed in ATHENA³⁴ using linear combination fitting. To examine the spatial distributions of arsenic valence states, chemical speciation maps were collected within the ROI and processed using SMAK.³⁵ To quantify the arsenic content relative to iron, the NIST SNRLXRF tool was used to generate calibration curves for the theoretical fluorescence intensities of Fe $K\alpha$ and As $K\alpha$ for a series of As-Pyrite compositions and for a series of arsenate–ferrihydrite compositions as previously demonstrated.³⁶ (See the [Supporting Information](#) for details.)

No thin section of the reacted fracture was prepared because the reacted wafer was too thin. Instead, we analyzed a section of another Eagle Ford sample from the same purchase and that had undergone a similar reactive flow experiment.¹³ This sample was cut so that the fracture face was perpendicular to the bedding, and the thin section was cut perpendicular to the fracture face to reveal the reacted and unreacted rock. The μXRF elemental maps of this thin section were acquired at the hard X-ray microprobe (13-ID-E) at the Advanced Photon Source (APS). A μXRF image of a 4 mm $\times$ 3.5 mm ROI was collected with a spatial resolution of 20 μm . All element maps were generated using LARCH.³⁷ (Detailed methods are in the [Supporting Information](#).)

2.4. xCT Imaging. The 3D structure and volumetric mineral distribution within the reacted fracture surface were characterized using X-ray computed tomography (xCT) at the APS beamline 13BMD. After the flow experiment, subcores (4

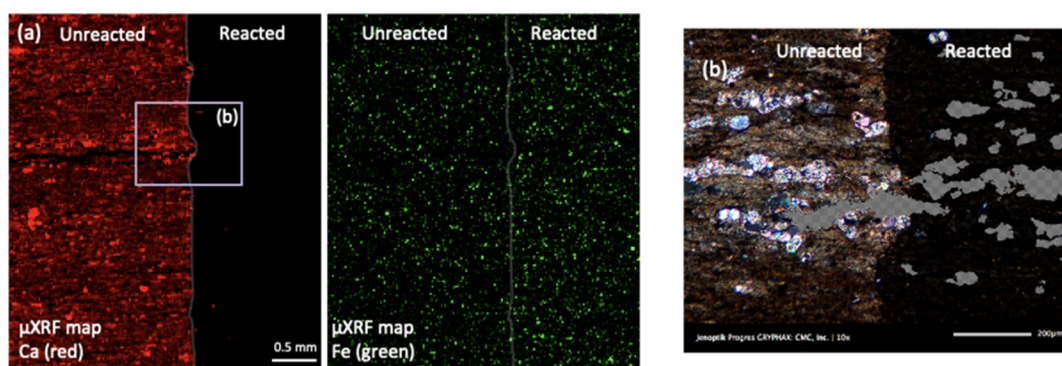


Figure 3. (a) μ XRF map of a cross-section of the altered and intact matrix after reaction. The gray line has been added to aid visualization of the boundary between the unreacted and reacted areas. The box represents the area imaged with a petrographic microscope. (b) Photomicrograph taken in the cross-polarized light mode, with image postprocessing to feature the pore space. The gray areas highlight big pores left behind by calcite dissolution, which also illustrates the orientation of the bedding as this thin section is cut perpendicular to the bedding.

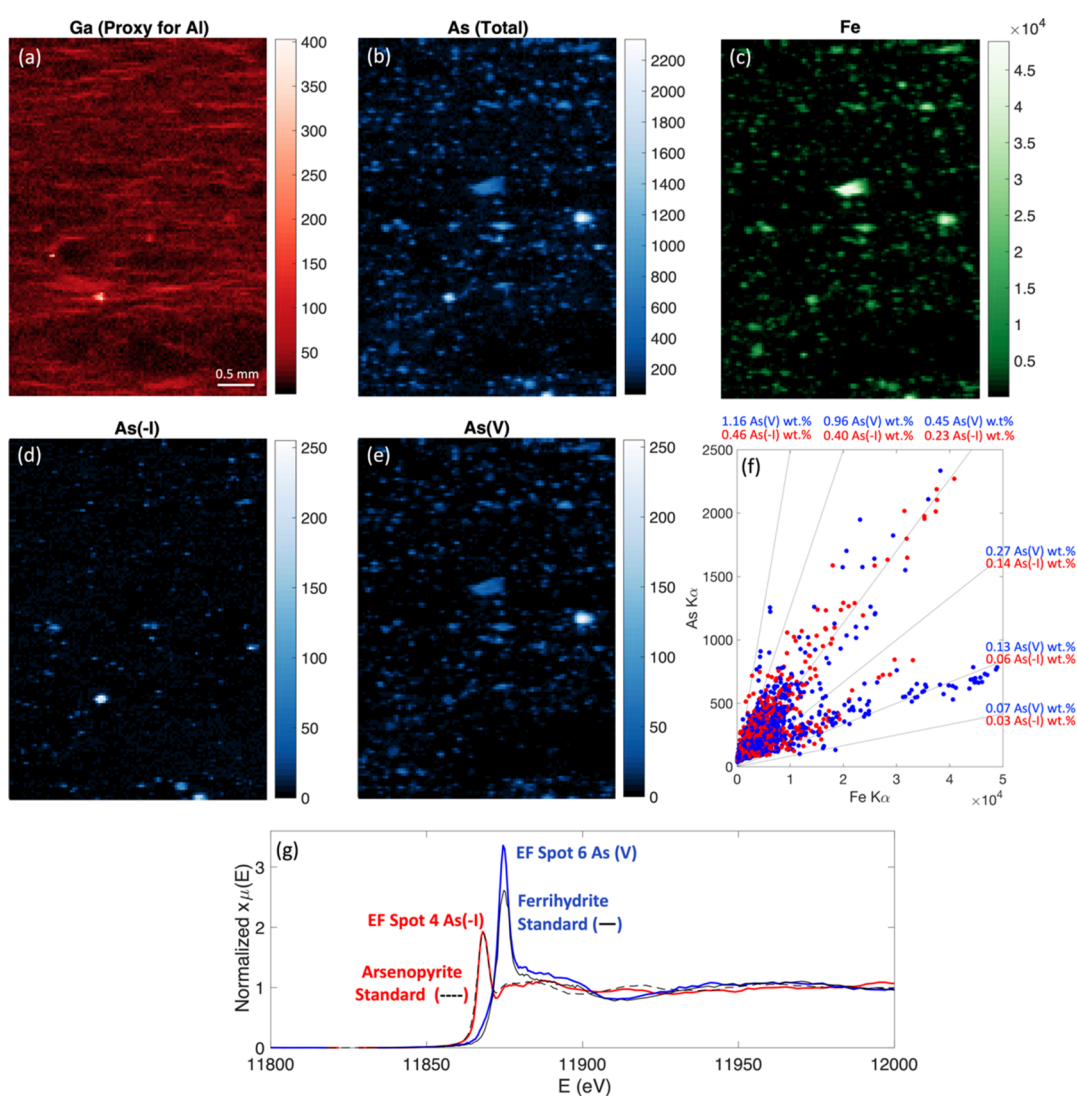


Figure 4. μ XRF characterizations of unreacted Eagle Ford shale (4 mm $\times$ 3.5 mm) of (a) gallium which serves as a proxy for aluminosilicates, (b) total arsenic, and (c) iron. Oxidation state maps of (d) As(-I) and (e) As(V). (f) As-Fe correlation plot showing blue As(V) and red As(-I) dominant pixels based on (d,e). Also shown in (f) are theoretical lines of As(-I) weight % in arsenopyrite and As(V) weight % associated with ferrihydrite from the SNRLXRF analyses. (g) Representative μ XANES spectra of two arsenic hotspots ("EF") compared with standard spectra of arsenopyrite and ferrihydrite.

mm diameter) of the reacted wafer, as illustrated in Figure 1b, were extracted in order to provide samples that could achieve

high-spatial resolution xCT images. The monochromatic X-ray beam was fixed at 30 keV, incoming beam intensity was

continuously monitored with a scintillation grid, and the sample-attenuated beam was focused with a 20 $\times$ objective onto a 5k $\times$ 7k CCD camera. The reconstructed tomograms had a voxel size of 4.6 μ m and were segmented to partition the solid phase and the pore space. The solid phase was further segmented by performing a binarization operation to distinguish calcite from the rest of the mineral matrix. This was possible because of the distinctly different X-ray attenuation strengths between the pore space and the solid phase and between calcite and clays, quartz, and metal sulfides.³⁸ Porosity and surface area estimates from the segmented images based on two thresholding methods (Otsu's method and visual inspection) are reported as a range, to quantify uncertainties associated with image processing. (xCT image processing is detailed in the Supporting Information.)

3. RESULTS AND DISCUSSION

3.1. Textural Changes and Implications for Reactive Transport. The xCT images of the postreaction fracture surface (Figure 2a,b) clearly show the formation of a porous altered layer consisting of minerals that were not removed by acid erosion. Analyses of the segmented xCT images (Figure 2c) indicate an average porosity of 63–68% for the altered layer depending on the thresholding method, in contrast to 1–2% for the unreacted shale matrix. The amount of new porosity in the altered layer is comparable to the calcite volume fraction in the unreacted shale, which is on average 63–70%. The xCT images also indicate that a minor amount of calcite remains in the altered layer and that this residual calcite is predominantly encased by the nonreactive mineral phases. The surface area, measured by the interfacial area between the solid phase and the pore space, within the altered layer is $\sim$ 32,000–32,500 m²/m³, which is more than 1 order of magnitude higher than that of the unreacted shale, 1500–2900 m²/m³. This difference is substantially larger than variations caused by image resolution.³⁹

The interface between the altered layer and the unreacted shale is the reaction front, which is sharp and shows clear topographic variations on the xCT images. The thickness of the altered layer, that is, the distance between the fracture surface and the reaction front, varies between 200 and 500 μ m. It should be noted that in the xCT images, there is no evident mineralogical difference between the hills and the valleys in the reaction front topography, which suggests that the variable thickness, that is, channeling within the altered layer, is likely caused by local perturbations. This is consistent with literature reports of linkages between hydrodynamic processes and textural or mineralogical heterogeneity at small scales.⁴⁰

When these textural changes occur because of acid perturbations in subsurface fractures, they will produce complex effects on reactive transport processes. The impact of the altered layer on flow rate has only recently been considered in simulation studies,¹² but previous flow experiments⁴¹ and modeling studies⁴² suggested that permeability evolution within a porous region may not follow standard power law relationships. Furthermore, two-phase flow dynamics are expected to be impacted because of mechanisms such as droplet fragmentation following pore structure alteration⁴³ and the wettability change, for example, calcite is a less-wetting surface compared to clays, quartz, and sulfides.^{44,45} It has also been reported that under compressive

stress, the altered layer may collapse and result in fracture sealing.^{13,46}

3.2. Mineralogical Characterization, Arsenic Accessibility, and Implications for Potential Mobilization. The μ XRF map and the petrographic image of the reacted Eagle Ford shale sample show that calcium is almost completely depleted in the altered layer (Figure 3a) and large pore spaces have emerged in the reacted area (Figure 3b). This supports the conclusions drawn from the xCT analysis that calcite had dissolved and porosity was created. PXRD patterns obtained from the altered layer further support this observation, with calcite diffraction peaks no longer present (see the Supporting Information for postreaction PXRD patterns). In contrast to calcium, iron remains in the altered layer with intensity that is apparently identical to that of the unreacted matrix (Figure 3a), which suggests that the iron-bearing minerals were not dissolved. The μ XRF maps of the unreacted sample highlight that arsenic-rich mineral phases co-occur with Fe-rich phases (Figure 4b,c), as is expected. In contrast, aluminosilicates (e.g., clay minerals) are uniformly distributed in the shale (Figure 4a). The strong co-occurrence of arsenic and iron is also evident in the As–Fe correlation plot (Figure 4f).

When acid-driven reactions of fracture surfaces happen in the subsurface, we expect that given the textural changes discussed above, accessibility and mobilization of the As- and Fe-bearing minerals can be enhanced in two ways. First, because of the porosity increase, the remaining structure is weakened and thus easier for the mineral grains to disaggregate and detach from the matrix. Calcite dissolution-induced release of pyrite particles in the Eagle Ford shale has been experimentally observed.¹⁸ Second, the increase in the surface area in the altered layer makes As- and Fe-bearing minerals more accessible, and this is expected to accelerate dissolution, oxidation, and desorption reactions that can transfer arsenic to the water. The extent to which these mobilization mechanisms are activated will depend on the speciation of arsenic and their weight percentages. Oxidation of the sulfur in pyrite can mobilize and oxidize As(-I). Oxidation can be enhanced as released minerals are transported to shallow groundwater formations where it is more oxidizing.²³ The desorption mechanism is most relevant to As(V).

In our work, the μ XANES point spectra collected at selected As–Fe hotspots were compared with mineral standard spectra and revealed that As(V) and As(-I) are the dominant oxidation states in our sample. Based on the mineral standard matches, we infer that As(V) is associated with iron-oxyhydroxides and As(-I) is associated with pyrite-like phases, which is consistent with what others have observed.^{19,47} To quantify the relative amounts of the two forms of As, the counts from each of the oxidation state maps (Figure 4d,e) were summed and the ratio of As(-I) to As(V) was determined to be approximately 0.6. To quantify the amounts of As in each of the iron-rich phases, the SNRLXRF-based theoretical analyses were used to generate the As wt % lines in Figure 4f. The As(-I) data were found to fall between 0.03 and 0.5 wt % of arsenic in pyrite. The As(V) data fell between 0.07 and 1.2 wt % of arsenic associated with ferrihydrite.

Our results show that arsenic accessibility and the potential for mobilization are increased by the development of the altered layer. Using the X-ray transmittance data, we were able to map the development of the altered layer quantitatively over time, as presented in the following section.

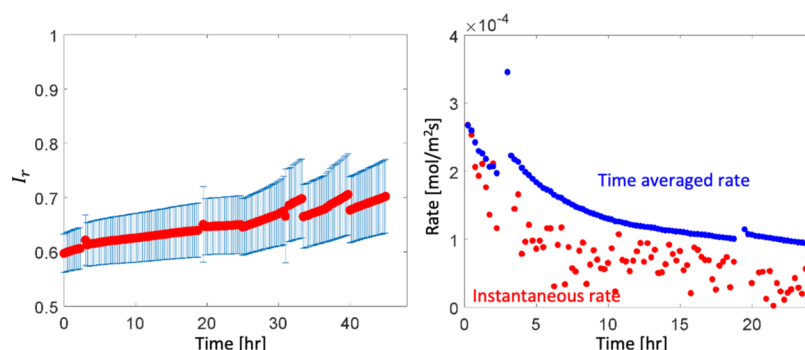


Figure 5. (a) Transmittance (I_r) measurements over time. The red dots are average values for the flow channel observation area outlined in Figure 1b and the blue bars are the standard deviations. (b) Calcite reaction rate calculated using measurements of two consecutive steps, that is, with a 15 min interval (instantaneous rate—red data points), and using the initial time step as the reference (time averaged rate—blue data points). Rates were calculated only for the first 24 h because of the reasons stated in Section 2.2.

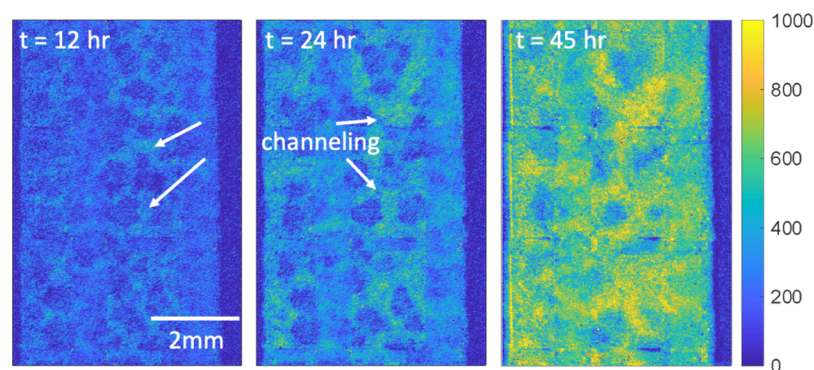


Figure 6. X-ray transmission images ($1.3 \mu\text{m}/\text{pixel}$) converted into depth of calcite dissolution, that is, thickness of the porous altered layer [μm] for the flow channel observation area in Figure 1b. The white arrows highlight the channeling in the altered layer.

3.3. Fracture Surface Erosion and Altered Layer Development. Figure 5a shows that the measured X-ray transmittance increased throughout the microfluidic flow experiment. This can be solely attributed to calcite dissolution given that the microfluidic cell design isolates changes occurring at a single fracture surface and minimizes rock mass loss due to mechanical effects, which is sometimes observed in fractured core flow experiments. The mineralogical characterization also confirmed that the noncalcite phase did not dissolve within the timeframe of the experiment.

This dataset was used to quantify altered layer thickness and estimate a fracture surface reaction rate. The transmission through the reacted sample can be written as

$$\ln(I_r) = A - L_w \mu_w - L_r \mu_{nc} f_{nc} - L_{AL} \mu_w (f_c + \phi) - (L_r - L_{AL})(\mu_c f_c + \mu_w \phi) \quad (3)$$

where L_{AL} is the thickness of the altered layer. This formulation implicitly assumes that the location of the original fracture surface does not change over time, which is confirmed by the xCT images (Figure 2a). Over time, the altered layer thickness increases and can be calculated as

$$L_{AL}(t) = \frac{\ln(I_r^t) - \ln(I_r^0)}{(\mu_c - \mu_w) f_c} \quad (4)$$

where the superscript t is time and the superscript 0 denotes the initial value. The attenuation coefficient for calcite and water under the incident energy conditions is 5.0 and 0.3745

cm^{-1} , respectively, according to the XCOM database.⁴⁸ A value of 67% was used for f_c based on the xCT image analysis.

Figure 6 shows 2D maps of the calculated altered layer thicknesses after 12, 24, and 45 h of reactive flow. The average thickness is $190 \mu\text{m}$ at hour 12, $300 \mu\text{m}$ at hour 24, and $580 \mu\text{m}$ at hour 45. The last value is in good agreement with the measurements derived from the xCT images (Figure 2). There are considerable spatial variations in the altered layer thickness, illustrating channelization of the reaction front below the fracture surface. The initial fine channels are clearly identifiable at hour 12. By hour 24, the channels grow into several wider paths separated by underlying hills where the altered layer is thin. After longer reactive flow, these hills shrink and the channels become less well-defined. The remaining hills correspond to the topographic hills of the reaction front in the xCT images.

The dissolution rate of the fast-reacting mineral has typically been used as an indicator of altered layer development and a modeling constraint. In our experiment, assuming the calcite surface area is the fracture surface area weighted by its volume fraction, the calcite reaction rate at a given time is

$$r(t) = \frac{L_{AL}(t_2) - L_{AL}(t_1)}{V_{m_c}(t_2 - t_1)} \quad (5)$$

where V_{m_c} is the molar volume of calcite and t_2 and t_1 are discrete time steps. The normalization of reaction rate used the nominal geometric surface area instead of the actual surface area, as the focus is to evaluate the change in the reactive flux over time.

The instantaneous reaction rate is calculated from two consecutive time steps, 15 min apart, whereas the time-averaged rate is calculated with the initial time step as the reference. The initial calcite dissolution rate spatially averaged over the channel is $\sim 3 \times 10^{-4}$ mol/m² s (Figure 5b). This rate is lower than that predicted by the kinetic rate law (approximately 0.089 mol/m² s for a pH of 2.2) because of the transport limitation given the large Peclet number (~ 500) of the system.⁴⁹ There is a clear decreasing trend in the calculated rates. The instantaneous reaction rate by hour 24 ($\sim 3 \times 10^{-5}$ mol/m² s) decreased by 1 order of magnitude, and the average reaction rate for the first 24 h is $\sim 9 \times 10^{-5}$ mol/m² s.

This trend is consistent with an increasingly diffusion-limited reaction as the thickness or depth of the altered layer increases. The diffusion-limited reaction rate can be estimated based on the diffusive flux across the fracture surface.⁴ For a porosity of 60% and an altered layer thickness of 100 μ m, the diffusion-limited reaction rate is in the order of 1×10^{-5} mol/m²s, which is comparable to the measured instantaneous reaction rate at hour 24 (detailed calculation is in the Supporting Information).

This study has demonstrated that the microfluidic cell can be coupled with synchrotron-based techniques for investigating acid erosion of fracture surfaces. With the rapid development of synchrotron-based characterization techniques and *in operando* capabilities, such systems are expected to provide more *in situ* observations that are otherwise impossible and thus to play an increasing role in advancing our fundamental understanding of rock–water interactions. The data collected in our study can be used to formulate future reactive transport models to improve our predictive understanding of altered layer development, permeability evolution, and arsenic release rate. Previous models developed for predicting altered layer development and fluid chemistry of major cations relied on integration of analytical representation of the transport process across the altered layer.⁴ This continuum approach does not resolve flow, mineral reactive surface area, and local fluid chemistry explicitly in the altered layer, which is critical for arsenic reactions. For instance, arsenopyrite dissolution was reported to increase with cation activity and arsenic desorption can be enhanced by the presence of carbonates.^{50,51} Even though pore-scale or multiscale 3D models have expanded our capability to predict local fluid chemistry and mineral accessibility and to fully account for processes that affect mobilization of trace elements in the matrix, their application has been limited to single-time snapshots.⁵² This is partly due to the lack of a suitable dataset for model development and validation. In our study, the parallel channel design and xCT images of the shale matrix provide necessary data for constructing a detailed 3D reactive transport model. The spatial and temporal evolution of the altered layer, along with the postreaction xCT image data, can be used to validate that the coupled chemical–physical processes responsible for the textural and mineralogical evolution are accurately described in the model. More importantly, the detailed characterizations of initial arsenic abundance and the oxidation state can be used in the validated model to predict As mobilization, following the acid erosion of the fracture surface.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.0c03736>.

Details of the microfluidic cell and X-ray attenuation setup, X-ray fluorescence data collection, powder X-ray diffraction analyses, xCT image segmentation, quantification of arsenic content, and calculation of diffusion-limited reaction rate (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work is supported by the National Science Foundation under Grant CMMI-0919140, NSF Grant CBET-1134397 to Princeton University, and Laboratory Directed Research and Development (LDRD) funding from Berkeley Lab, provided by the Director, Office of Science of the U.S. Department of Energy (DOE) under contract no. DE-AC02-05CH11231. It is also supported by Princeton Environmental Institute at Princeton University. Portions of this work were performed at the National Synchrotron Light Source under the U.S. DOE contract no. DE-AC02-98CH10886 and at the XFM (4-BM) Beamline of the NSLS II under the U.S. DOE contract no. DE-SC0012704 operated by Brookhaven National Lab. Portions of this work were performed at GeoSoilEnviroCARS (The University of Chicago, Sector 13), at the Advanced Photon Source, which is operated for the DOE Office of Science by Argonne National Laboratory under contract no. DE-AC02-06CH11357. GeoSoilEnviroCARS is supported by the National Science Foundation - Earth Sciences (EAR-1634415) and DOE GeoSciences (DE-FG02-94ER14466).

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